OPPORTUNITY BRIEF · SALES AND PRESALES

AI-integrated drone and ground robots for cleaning and predictive maintenance of solar panels.

Energy × Predictive maintenance × Drones and UAS · OS149 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
46 is the world moving	**43** can we play, can we win	**MEDIUM** 0 named in the evidence	**€7.8m** partial evidence · per year	**LATER** research and standards activi...	**L1** 5 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 3.0 published reference(s) in Energy, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity targets solar plant operators in dusty, high-temperature regions where panel performance degrades due to dust and heat. By integrating AI-driven drones and ground robots for real-time monitoring, predictive analytics, and intelligent cleaning, Orange can offer a managed service that enhances solar panel performance. The time is right as AI and robotics technologies have matured, and the need for reliable renewable energy integration is pressing.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	**€9.5m** €1.5m – €146m	**€7.8m** €1.2m – €120m	**€94k** €15k – €1.4m	partial
Observed: tendered contract value, annualised	**€172m** €172m – €460m	**€172m** €172m – €460m	**€2.1m** €2.1m – €5.5m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Energy (EU27_2020)	5,796 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Drones and UAS	2.4 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_ain2 · E_AI_TAR	2025
Annual value of one engagement	€299k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	1,341 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Artificial intelligence by NACE Rev. 2 activity series E_AI_TAR is used as a proxy for Drones and UAS: Autonomous machines — includes autonomous drones. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 1 declared geographies are outside the European reference data (IN); the estimate covers the European footprint only.

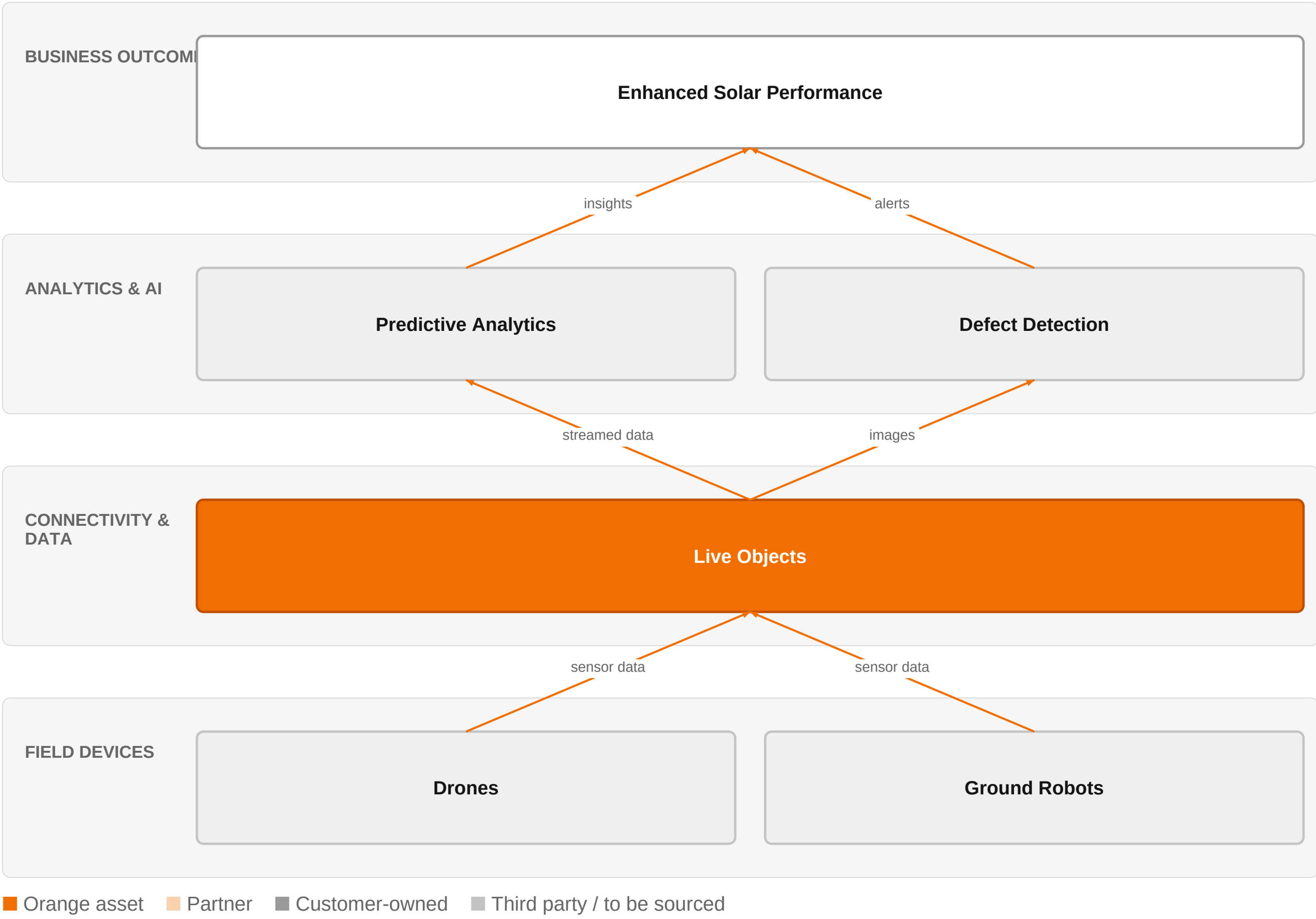
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€172m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 21 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 1 declared geographies are outside the European reference data (IN); the estimate covers the European footprint only.

The solution, in one picture

AI-Integrated Solar Panel Maintenance



This diagram shows how Orange's IoT platform connects drones and robots to AI analytics, enabling proactive maintenance that boosts solar panel output.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The market is shifting toward AI-integrated autonomous systems for solar panel maintenance, as evidenced by recent research combining real-time monitoring, predictive analytics, and intelligent cleaning. Deep learning models for defect detection in photovoltaic panels are advancing, enabling more efficient automated inspection. Additionally, the integration of AI and digital twin technology in manufacturing is setting a precedent for predictive maintenance approaches that can be applied to energy infrastructure.

`Sources: SIG-39681DB73044 openalex.org 2025-09-01; SIG-AA5215E67485 openalex.org 2025-09-26; SIG-0964410DC575 openalex.org 2025-12-17`

Who buys, and why now

The buyer is typically the operations or maintenance director at a solar plant operator, especially in regions with high dust or heat. The pain is operational: dust accumulation and surface heating cause measurable performance degradation, and manual cleaning and inspection are costly and infrequent. The trigger is often a drop in energy output or a failed inspection that reveals widespread defects, prompting a search for more efficient, predictive solutions.

Why it is hot now — every claim bound to a dated source

- AI-integrated autonomous robotics combine real-time monitoring, predictive analytics, and intelligent cleaning to enhance solar panel performance. [SIG-39681DB73044]

- An AI-integrated autonomous robotic system combines CNN-LSTM-based fault detection and reinforcement learning-driven robotic cleaning to enhance solar panel performance. [SIG-39681DB73044]
- AI-integrated autonomous robotics combine real-time monitoring, predictive analytics, and intelligent cleaning for enhanced solar panel performance. [SIG-39681DB73044]
- The proposed system integrates edge computing for real-time monitoring and intelligent cleaning to combat performance degradation from dust and heat. [SIG-39681DB73044]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its IoT experts and Orange Group researchers to design an AI-integrated monitoring and cleaning system. The core would leverage Live Objects for secure device connectivity and data management, with ISO 27001 and SecNumCloud certifications ensuring trust and sovereignty. The engagement would start with a pilot on a customer's solar site, deploying drones and ground robots equipped with sensors and cameras, connected via Live Objects, and using AI models for defect detection and cleaning optimization.

Why Orange can win

Orange can win by combining its IoT expertise, secure cloud infrastructure, and research capabilities to deliver a trusted, end-to-end managed service. The certifications (ISO 27001, SecNumCloud) provide a sovereign edge that pure tech vendors may lack. However, the assets are thin: Orange does not have its own drone or robotics hardware, so it would need to partner or integrate third-party devices, which could weaken its value proposition.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer addresses the use case but not this technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 5.7; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
Airbus Protect	Security specialist	sells Drones and UAS	structural	—
Dedrone by Axon	Security specialist	sells Drones and UAS; competes in OX: Smart Industries	structural	—
HENSOLDT	Security specialist	sells Drones and UAS	structural	—
Leonardo	Security specialist	sells Drones and UAS	structural	—
Thales	Security specialist	sells Drones and UAS	structural	—
Capgemini	Global systems integrator	active in Energy; competes in OX: Smart Industries	structural	—
Telefonica Tech	Telecom operator peer	active in Energy	structural	—
Honeywell	Industrial / OT specialist	active in Energy; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

1. How much energy output do you lose annually due to dust or soiling on your panels?

2. What is your current cleaning frequency and cost per cleaning cycle?
3. Do you currently use any drone or robotic inspection on your solar sites? If so, what data do you collect and how is it used?
4. What is your biggest pain point: cleaning efficiency, defect detection, or maintenance planning?
5. Are you operating in a region with high dust or high temperatures that accelerates performance degradation?
6. What are your data sovereignty or security requirements for any monitoring solution?

Objections you will meet

They will say	You can answer
We already have a cleaning contractor and do manual inspections.	That's true, and manual methods work at small scale. But as your plant grows, manual cleaning becomes less frequent and inspections miss early defects. Our AI-integrated approach combines real-time monitoring with predictive analytics, so you catch issues before they cause significant energy loss.
Drones and robots are expensive and unproven in our environment.	You're right that the technology is still evolving. That's why we propose a pilot on a subset of your panels to measure the actual performance improvement and ROI. Our IoT platform and secure cloud ensure the data is reliable, and we can adjust the solution based on your specific conditions.
We're concerned about data security and sovereignty.	That's a valid concern. Orange's solution is built on our Live Objects platform, which is ISO 27001 certified, and we offer SecNumCloud-certified hosting for the highest level of sovereignty. Your data stays under your control and within your jurisdiction.

Risks and unknowns

The main risk is that Orange lacks proprietary drone and robotics technology, making it reliant on partners. Additionally, the technology is still in research phase (time horizon: later), so commercial readiness is unproven. Regulatory approval for drone operations over solar plants may vary by region, and the business case depends on the cost of cleaning vs. energy loss, which is not fully quantified.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a solar farm operator, outlining how Orange's IoT experts and Live Objects can provide the secure, scalable data platform to integrate with AI-integrated robotic cleaning and predictive maintenance solutions, emphasizing ISO 27001 and SecNumCloud certifications for data security.
Sales	In a conversation with a renewable energy operator's operations director, say: 'Orange's IoT expertise and secure cloud solutions, such as Live Objects, can support the data management and connectivity needs of AI-driven robotic systems for solar panel maintenance, helping you optimize performance and reduce downtime.'
Strategist / Innovator	Commission a deep-dive brief on AI-integrated autonomous robotics for solar panel cleaning and predictive maintenance, leveraging Orange Group researchers and IoT experts to assess technical feasibility and market readiness, and to identify potential partnerships in the energy sector.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-B81FDE60EE21	2026-05-01	cordis.europa.eu	1	Smart Upgraded iNverter architectures for Versatile and flexible Integration of PV-Based systEms
SIG-0964410DC575	2025-12-17	openalex.org	1	Generative and Predictive AI for digital twin systems in manufacturing
SIG-AA5215E67485	2025-09-26	openalex.org	1	Optimized YOLO based model for photovoltaic defect detection in electroluminescence images
SIG-39681DB73044	2025-09-01	openalex.org	1	AI-Integrated autonomous robotics for solar panel cleaning and predictive maintenance using dro...
SIG-D29A41589B0F	2026-07-25	arxiv.org	3	Movable-Antenna Assisted Energy Minimization in UAV-Enabled Mobile Edge Computing Systems

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:24+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.